



ZEWAIL CITY
DSAI 103

PROJECT PHASE (1): EBAY PRODUCTS SCRAPER & NETWORK ANALYSIS

Data Collection, Preprocessing, and Initial
Network Insights for Python Books

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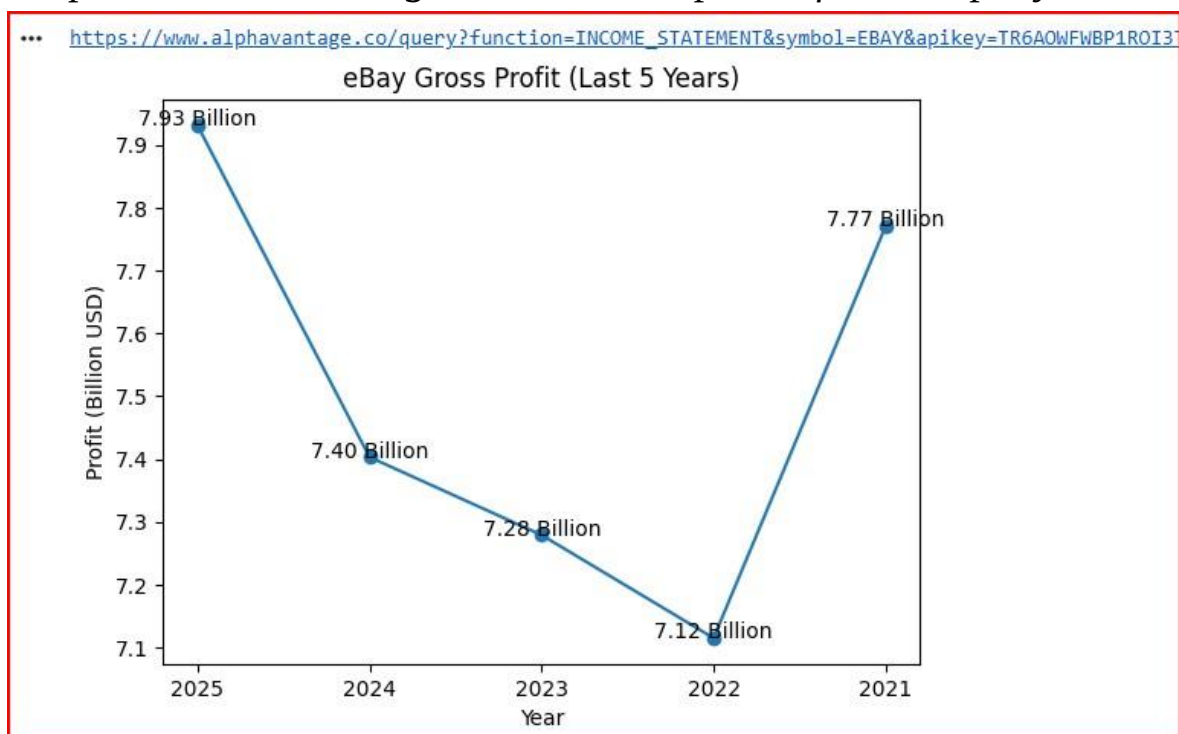
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1- Introduction & Data Collection:

- In this project, we aimed to build a comprehensive data pipeline for Python programming books available on eBay. The goal is to collect, clean, and analyze the relationships between products using network science.

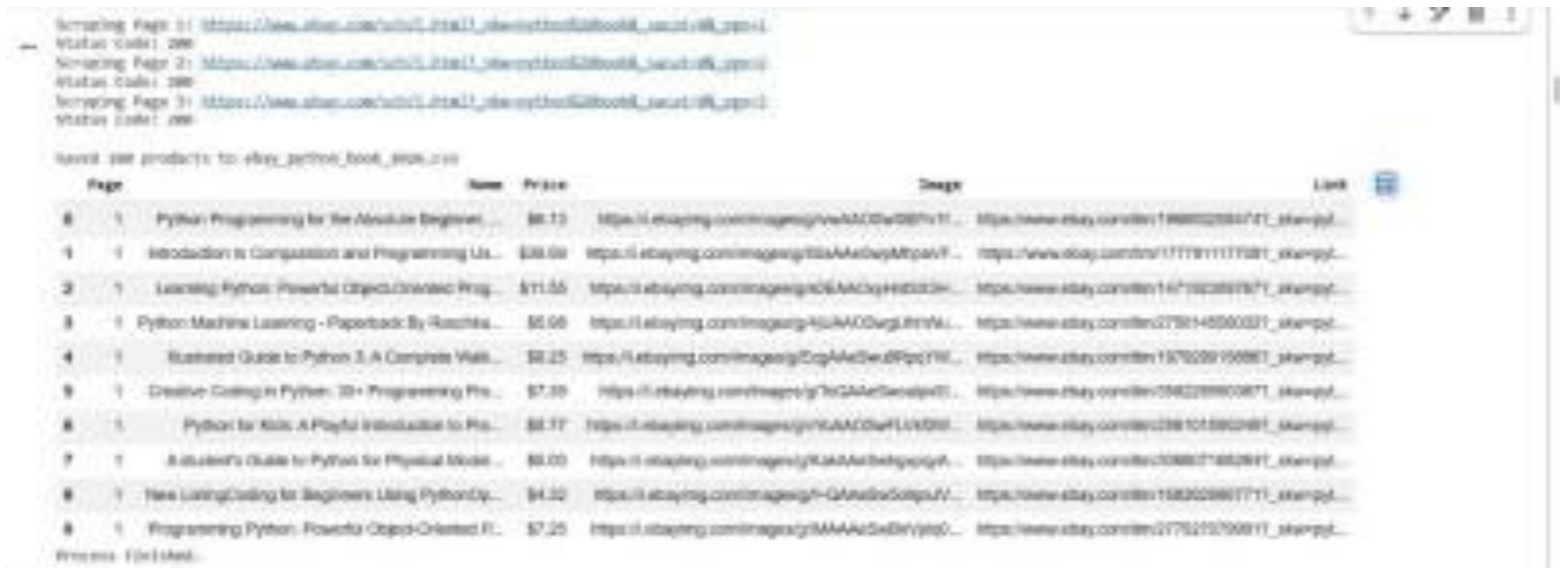
2.1-introduction about EBAY profits:

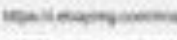
- We use extra methods from alpha vantage to extract the company profits among the previous 5 years and graph it by using matplotlib and also we get the annual reports of the company



2.2- Data Collection (The 3 Methods):

- **Beautiful Soup:** Used to crawl 181 static product listings while pandas was utilized to organize the extracted data into a structured Data Frame and export it to a CSV file for further analysis.



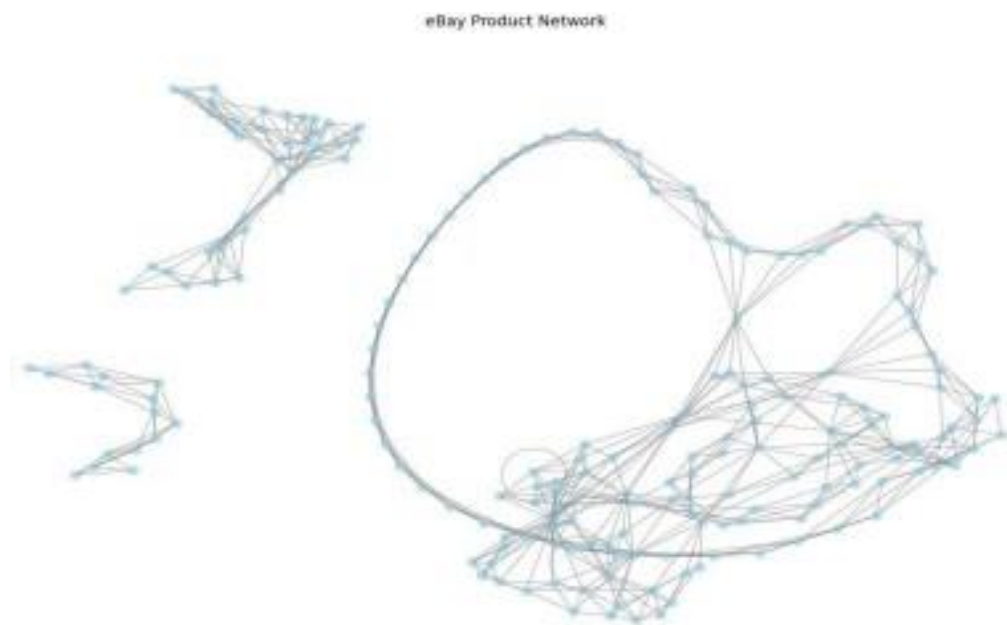
Page	Name	Price	Image	Link
1	Python Programming for the Absolute Beginner...	\$8.13		https://www.ebay.com/itm/188822881771?_st=pyt...
1	Introduction to Computation and Programming Us...	\$8.99		https://www.ebay.com/itm/177781117781?_st=pyt...
1	Learning Python: Powerful Object-Oriented Prog...	\$11.55		https://www.ebay.com/itm/141732289767?_st=pyt...
1	Python Machine Learning - Paperback By Raschka	\$2.99		https://www.ebay.com/itm/273214550032?_st=pyt...
1	Automated Guide to Python 3: A Complete Walk...	\$8.25		https://www.ebay.com/itm/137629815081?_st=pyt...
1	Creative Coding in Python: 30+ Programming Pro...	\$7.39		https://www.ebay.com/itm/258228860871?_st=pyt...
1	Python for Kids: A Playful Introduction to Pro...	\$8.77		https://www.ebay.com/itm/238101592981?_st=pyt...
1	A student's guide to Python for Physical Model...	\$8.00		https://www.ebay.com/itm/208827482811?_st=pyt...
1	New Learning/Doing for Beginners Using Python/Py...	\$4.20		https://www.ebay.com/itm/158202861771?_st=pyt...
1	Programming Python: Powerful Object-Oriented P...	\$7.25		https://www.ebay.com/itm/217521570811?_st=pyt...

- **Selenium:** Used to scrape 50 dynamic listings that require JavaScript rendering.
- **eBay API:** Integrated to pull structured data for higher accuracy. Data was retrieved using SerpApi, converted into a pandas Datagram, and exported as a CSV file for analysis

3- Data Preprocessing:

- **Merging:** Combined data from CSV files into a master dataset.
- **Price Cleaning:** Converted price strings (e.g., "\$10.50") into numerical floats for calculation.
- **Categorization:** Categorized books into "Beginner", "Data Science", "Web Dev", and "General" based on keyword analysis.

4- Network Analysis:



- **Graph Construction:** We used **NetworkX** to build a product relationship graph. Edges were created based on shared categories.
- **Key Nodes (Centrality):** Based on **Betweenness Centrality**, we identified the most influential books in the network. These books act as "hubs" that connect different programming sub-fields.
- **Insights:** The visualization shows clear clusters for "Data Science" and "Beginners" books, indicating high relevance within these groups.

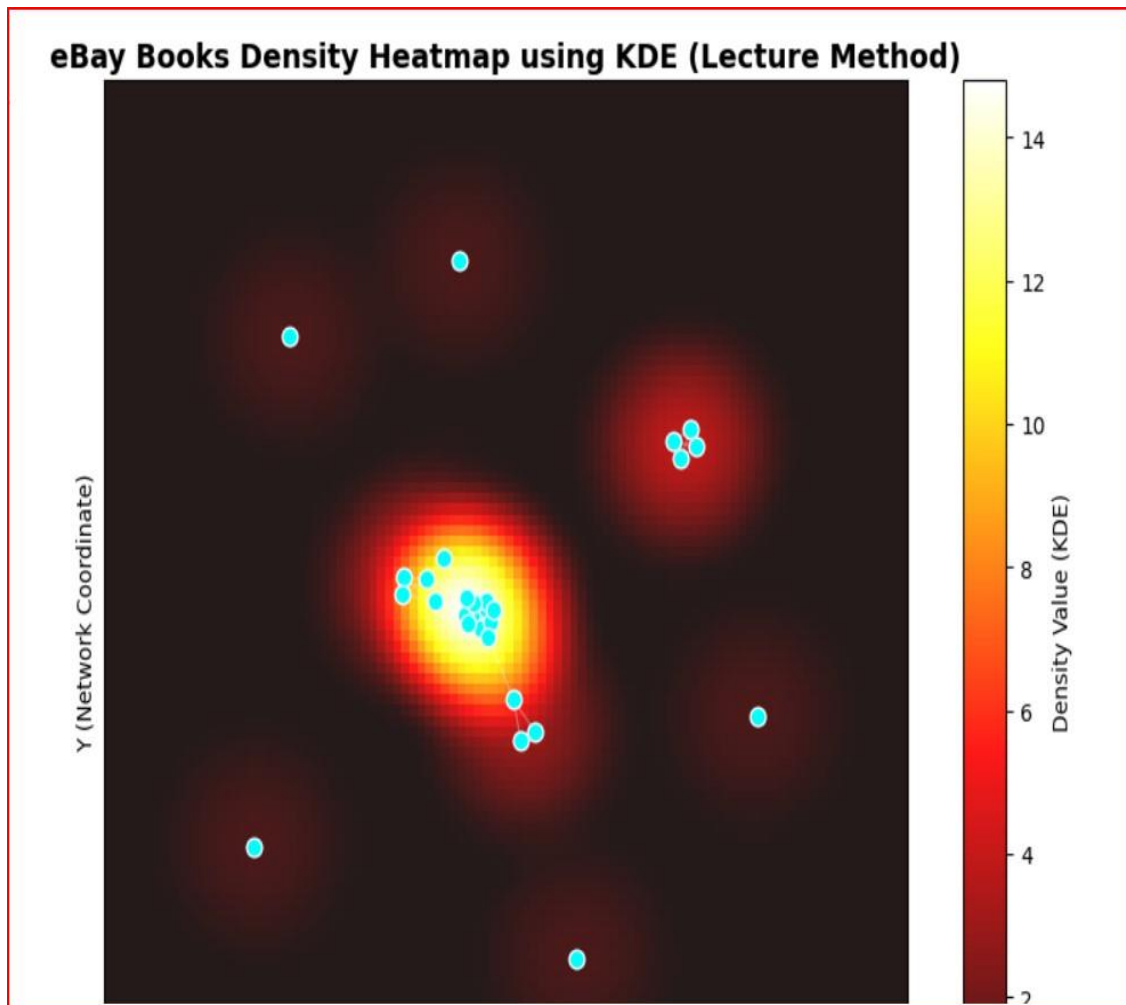
5- Conclusion of phase 1:

Phase 1 successfully established the data infrastructure. We achieved a robust scraping mechanism and a meaningful network visualization. **Note:** As discussed, advanced Heatmaps and further community detection will be explored in **Phase 2**.

6- Heat maps:

- A KDE-based heatmap was used to visualize the density of nodes in the book network. The graph was built from 40 books, where connections represent shared words between titles. Using node positions from the spring layout, density was calculated based on the distance between nodes within a fixed radius.

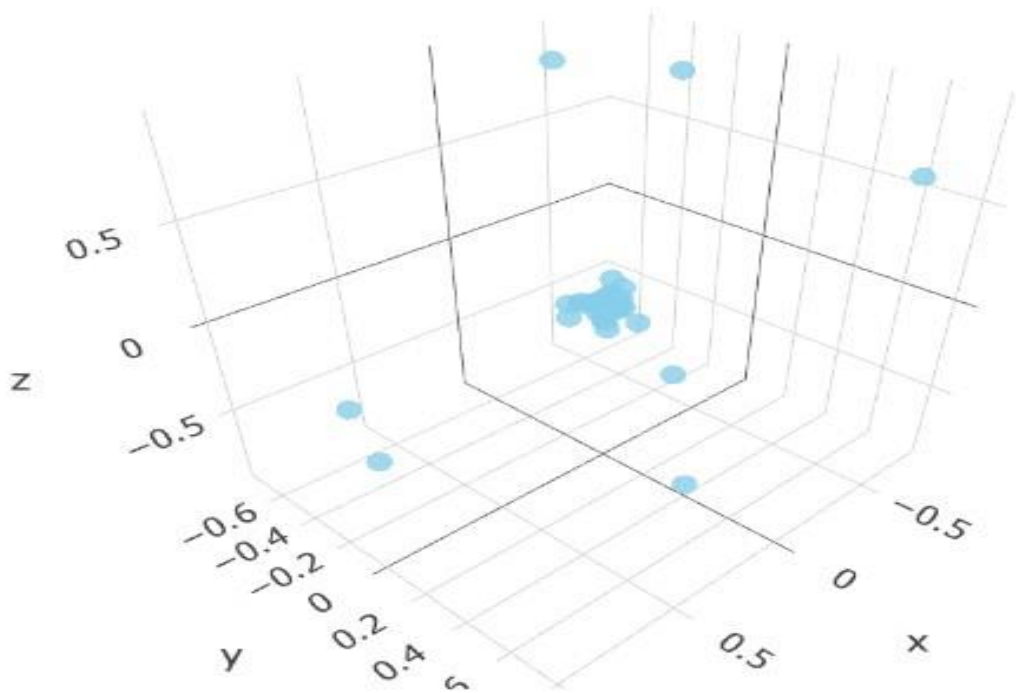
- The heatmap clearly shows a strong indicating highly related books, along with smaller clusters and some isolated nodes. This helps in identifying patterns, similarities, and the overall structure of the network.



7-Point Cloud Analysis using 3D representations:

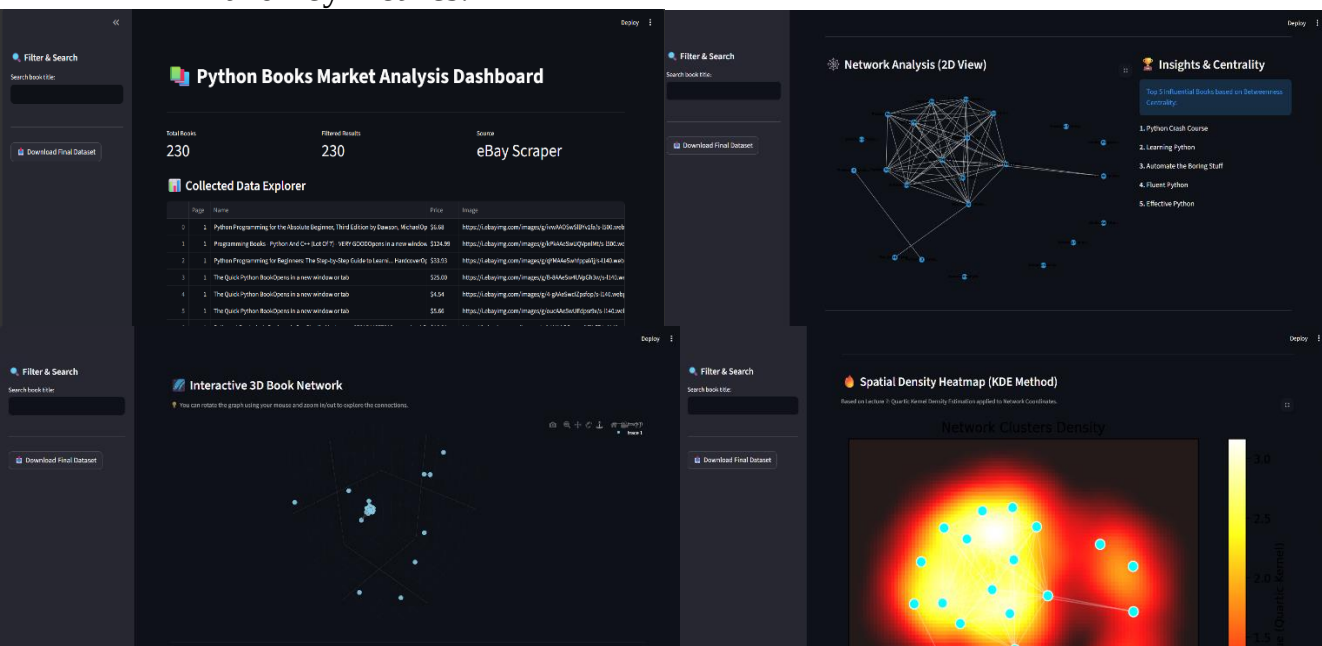
- As shown in the figure, a 3D point cloud representation of the books relationship network was created using Plotly. Each point represents a book positioned in 3D space, while the distribution of points reflects the underlying structure of the network.
- The visualization reveals a noticeable concentration of nodes near the center, indicating a group of closely related books. In contrast, a few nodes appear more distant, suggesting weaker or less common relationships within the dataset.
- This 3D representation enhances the understanding of the network by making the relative distances between books more visually apparent.

Phase 2: 3D Point Cloud - Books Relationship Network



8-Making the Final GUI

- The application is an interactive dashboard built using Streamlit to explore and analyze a dataset of Python books collected from eBay. It enables users to filter data, visualize relationships using NetworkX, and extract insights through dynamic tables and key metrics.



9-Conclusion of phase 2 :

- Phase 2 enhanced the project by adding advanced visualizations such as heatmaps and 3D point cloud representations, which provided clearer insights into the structure and relationships within the dataset. Additionally, the interactive GUI improved usability by allowing users to explore, filter, and analyze the data dynamically. Overall, this phase made the analysis more intuitive, interactive, and insightful.